KDK College of Engineering Nagpur

6th Sem BTech(IT)

DBMS

Database System Structure & architecture

Database Management System (DBMS) is software that allows access to data stored in a database and provides an easy and effective method of –

- Defining the information.
- Storing the information.
- Manipulating the information.
- Protecting the information from system crashes or data theft.
- Differentiating access permissions for different users.

Used by the computer system on which the database system runs.

- Database systems can be centralized, or client-server.
- A database system is divided into modules that deal with different responsibilities of the overall system.

- Primary goal:- retrieving information from and storing new information into the database.

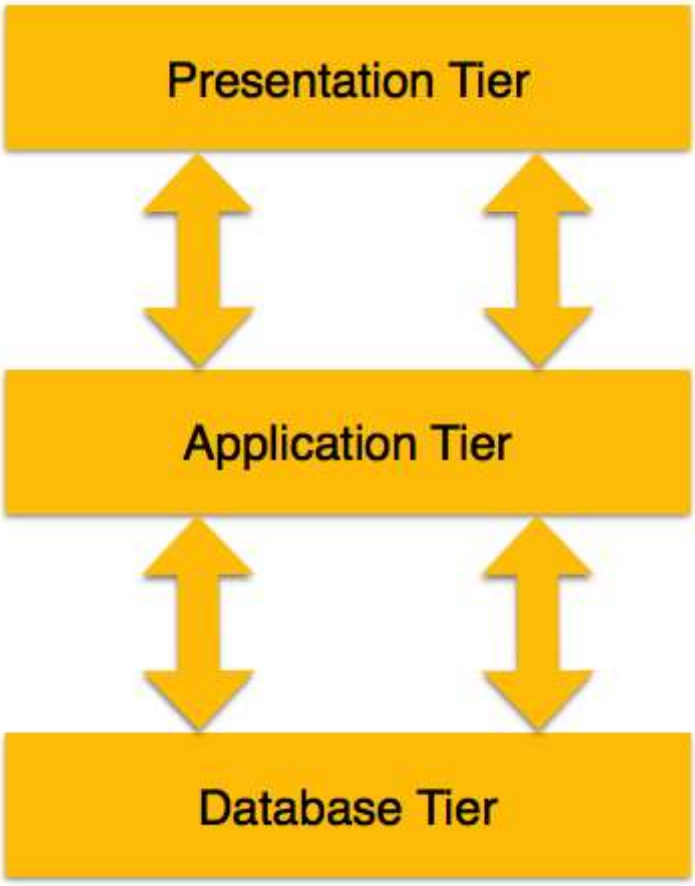
The design of a DBMS depends on its architecture. It can be centralized or decentralized or hierarchical. The architecture of a DBMS can be seen as either single tier or multi-tier. An n-tier architecture divides the whole system into related but independent **n** modules, which can be independently modified, altered, changed, or replaced.

In 1-tier architecture, the DBMS is the only entity where the user directly sits on the DBMS and uses it. Any changes done here will directly be done on the DBMS itself. It does not provide handy tools for end-users. Database designers and programmers normally prefer to use single-tier architecture.

If the architecture of DBMS **is 2-tier**, then it must have an application through which the DBMS can be accessed. Programmers use 2-tier architecture where they access the DBMS by means of an application. Here the application tier is entirely independent of the database in terms of operation, design, and programming.

3-tier Architecture

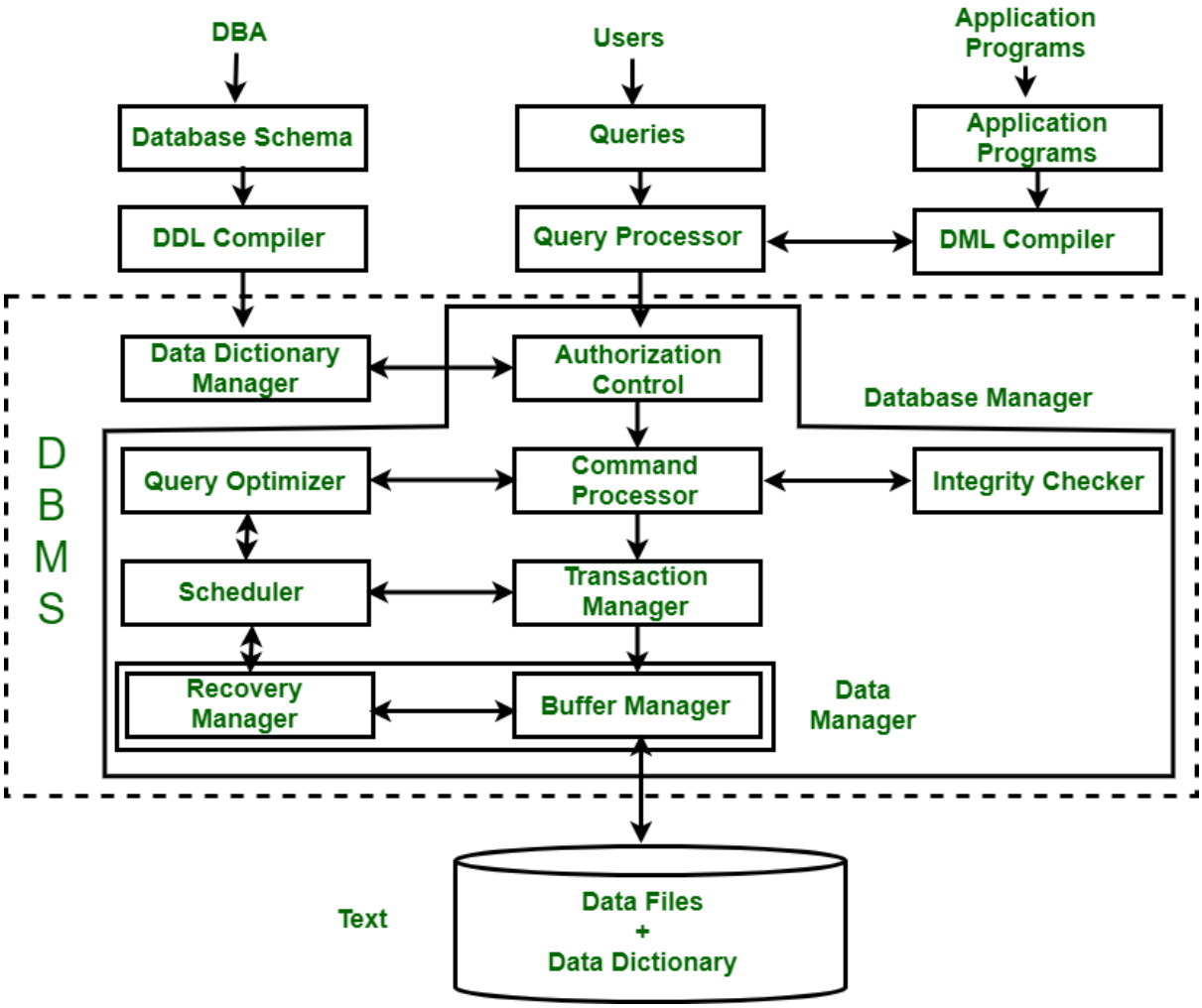
A 3-tier architecture separates its tiers from each other based on the complexity of the users and how they use the data present in the database. It is the most widely used architecture to design a DBMS.

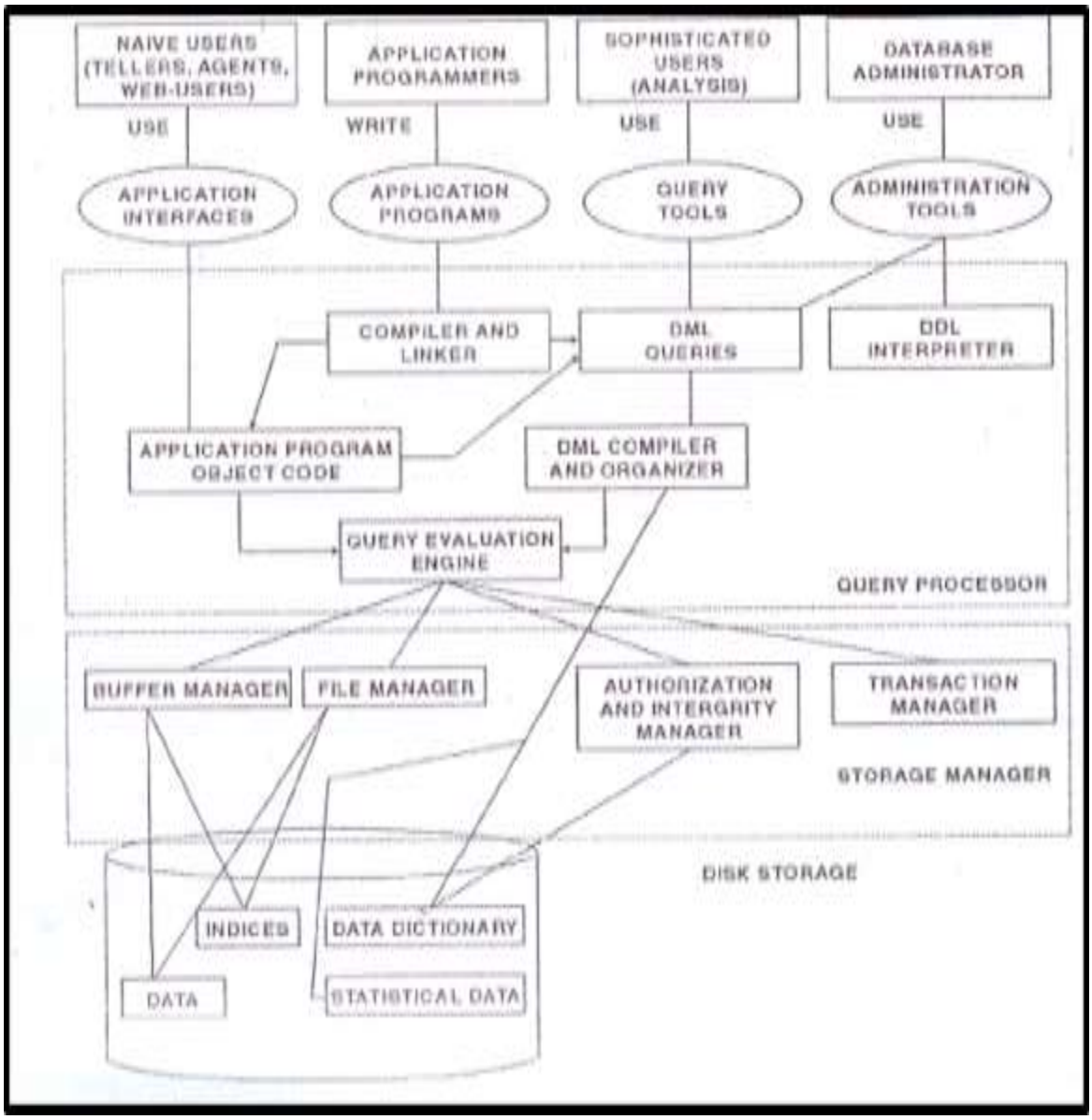


- **Database (Data) Tier** − At this tier, the database resides along with its query processing languages. We also have the relations that define the data and their constraints at this level.
- **Application (Middle) Tier** − At this tier reside the application server and the programs that access the database. For a user, this application tier presents an abstracted view of the database. End-users are unaware of any existence of the database beyond the application. At the other end, the database tier is not aware of any other user beyond the application tier. Hence, the application layer sits in the middle and acts as a mediator between the end-user and the database.
- **User (Presentation) Tier** − End-users operate on this tier and they know nothing about any existence of the database beyond this layer. At this layer, multiple views of the database can be provided by the application. All views are generated by applications that reside in the application tier.

Multiple-tier database architecture is highly modifiable, as almost all its components are independent and can be changed independently.

The database system is divided into functional components: User,interface,Query Processor, Storage Manager, and Disk Storage. These are explained as following below.


DBA
Users
Application Programs
Database Schema
Queries
Application Programs
DDL Compiler
Query Processor
DML Compiler
Data Dictionary Manager
Authorization Control
Database Manager
D B M S
Query Optimizer
Command Processor
Integrity Checker
Scheduler
Transaction Manager
Recovery Manager
Buffer Manager
Data Manager
Text
Data Files + Data Dictionary



Query Processor: It interprets the requests (queries) received from end user via an application program into instructions. It also executes the user request which is received from the DML compiler.

Query Processor contains the following components –

- **DML Compiler:** It processes the DML statements into low level instruction (machine language), so that they can be executed.
- **DDL Interpreter:** It processes the DDL statements into a set of table containing meta data (data about data).
- **Embedded DML Pre-compiler:** It processes DML statements embedded in an application program into procedural calls.
- **Query Optimizer:** It executes the instruction generated by DML Compiler.

2. Storage Manager: Storage Manager is a program that provides an interface between the data stored in the database and the queries received. It is also known as Database Control System. It maintains the consistency and integrity of the database by applying the

constraints and executing the DCL statements. It is responsible for updating, storing, deleting, and retrieving data in the database.
It contains the following components –

- **Authorization Manager:** It ensures role-based access control, i.e,. checks whether the particular person is privileged to perform the requested operation or not.

- **Integrity Manager:** It checks the integrity constraints when the database is modified.

- **Transaction Manager:** It controls concurrent access by performing the operations in a scheduled way that it receives the transaction. Thus, it ensures that the database remains in the consistent state before and after the execution of a transaction.

- **File Manager:** It manages the file space and the data structure used to represent information in the database.

- **Buffer Manager:** It is responsible for cache memory and the transfer of data between the secondary storage and main memory.

3. Disk Storage: It contains the following components –

- **Data Files:** It stores the data.

- **Data Dictionary:** It contains the information about the structure of any database object. It is the repository of information that governs the metadata.
- **Indices:** It provides faster retrieval of data item.

SYSTEM USERS

four different types of database-system users,

1. **Application programmers**
1. **Sophisticated users and Specialized users**:-
1. **Database Administrator**
1. **Naive users**

1. **Application programmers**

- Computer professionals and write application programs.
- They interact with the system through DML calls.
- DML calls are embedded in a program written in a host language (for example PHP, python, c).
- These programs are commonly referred to as application programs.

2. Sophisticated users and Specialized users:-

- Interact with the system without writing programs.
- They request in a database query language as query form.
- Each query is submitted to a query processor.
- Query processor is to break down DML statements into instructions that the storage manager understands.
- They work as an analyst.
- Specialized users are sophisticated users who write specialized database applications

like computer-aided design Systems, knowledgebase, and expert systems, etc.
• The user uses complex data types (for example, graphics data and audio data).

(3) **Database Administrator (DBA)**
• DBA has central control over the system.
• Responsible for following functions:
(i) Schema Design and Maintenance,
(ii) Physical Schema and Organization Modification,
(iii) Authorization and Security,
(iv) Integrity Constraint Specification,
(v) Recovery from Failure,
(vi) Database Upgradation

(4) Naive users
• **Naive users** are unsophisticated users.
• Interact with the system by the permanent application programs that have been written previously.
• For example, the clerk at the ticket booking window, he uses an application program to do his job of making reservations for a passenger.

Query Processor:
• A query processor helps the database system simplify and facilitate data access.
• System users are not required to know physical details of the implementation of the system.
• Quick processing of updates.
• Queries are written in a nonprocedural language, at the logical "level.
• Results are stored into an efficient sequence of operations at the physical level.

Query Processor Components: -
These components are used in evaluating DDL and DML queries. These are:-
(i) DDL Interpreter
• Interprets DDL statements and store the definitions in the data dictionary, as metadata.

(ii) DML Queries
• Sophisticated or Specialized user's requests in a database query language.
• Each query is submitted to a query tools
• Query tools break down it into DML queries and low-level instructions.
• These are sent to the DML compiler and Organizer, and the query evaluation engine for further processing.

(iii) DML Compiler and Organizer
• DML calls are usually started by a special character/code so that the appropriate code can be generated.
• A special preprocessor, called the DML precompiles, converts the DML statement to normal procedure calls in the host language.
• The resulting program is then run through the host-language compiler, which generates appropriate object code (a set of low-level instructions that can be used by query evaluation engine.)

(iv) Application Program Object Code

- It converts DML statements embedded in an application program to normal procedure calls in the host language.
- These pre-compilers consult the DML compiler to generate the appropriate code.

(v) Compiler and Linker: -

- Application programmer writes program application.
- The source codes compiled by the compiler and linker-linked application program object code to DML queries and send to query evolution engine.

(iv) Query evaluation Engine: -

- A query can usually be translated into any of a number of alternative evaluation plans that all give the same result.
- The DML compiler also performs query optimization, that is, it picks the lowest cost evaluation plan from among the alternatives.
- This component is responsible for interpreting and executing the SQL query.
- It contains three major components

Compiler - builds a data structure from the SQL statement and then does semantic checking on the query such as whether the table exists, field ‘exists, etc.

Optimizer - transforms the initial query plan (data structure created by the compiler), into the sequence of operations usually pipelined together to achieve fast execution.

It refers to the metadata (dictionary) and statistical information stored about the data to decide which sequence of operations is likely to be faster and based on that it creates the optimal query plan. Both cost and rule-based optimizers are used.

- **Execution Engine -** executes each step in the query plan chosen by Optimizer.

It interacts with the relation engine to retrieve and store records.

Storage Manager:

- A storage manager is a program module that provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system.
- The storage manager is responsible for the interaction with the file manager.
- The raw data are stored on the disk using the file system, which is usually provided by a conventional operating system.
- The storage manager translates the various DML statements into low-level file system commands.
- The storage manager is responsible for storing, retrieving, and updating data in the database.
- A large amount of storage space is required for storing corporate databases (which may range from hundreds to gigabytes to terabytes of data) and to manage this storage manager is required.
- Data are to move between disk storage and main memory as per requirement because the main memory of the computer cannot store this much information.

The storage manager components–

(i) File Manager

- It manages disk space allocation and the data structures used to store the data.
- File manager maps disk pages of the file to the memory pages at the disk in physical form and does the actual disk I/O operations in case of major faults generated "by buffer manager module.

(ii) Buffer Manager

- Buffer manager responsible for loading pages(fetching) from disk to main memory and to managing the buffer pool based on Least Recently Used (LRU) algorithm and deciding the caching strategy suitable for the application.
- It is a critical part of the database system, it enables the database to handle data sizes that are much large than the size of main memory for this has a special-purpose allocator for storing control information, which is transient.
- Buffer pool is the memory space used by buffer manager to cache disk pages associated with records, index information, Metadata information.
- Some database systems have space limits at the individual level and some at the global level for buffer pool size.

(iii) Transaction Manager

- The transaction manager creates transaction objects and manages their atomicity and durability.
- Applications request the creation of a transaction object by calling the transactions manager's begin Transaction method.
- When a resource manager first participates in a transaction, it calls the Enlist method to enlist in the transaction.
- The transaction manager tracks all the resource managers who enlist in the transaction.
- It ensures that the database remains in a consistent (correct) state despite system failures and that concurrent transaction executions proceed without conflict.

One of the following three results can occur:

1. The application either commits or aborts the transaction.
2 . A resource manager aborts the transaction.
3. A failure occurs.

(iv) Authorization and Integrity Manager

- This manager is responsible for granting access to the database or portions thereof only to authorized users and preventing access to unauthorized users.
- It tests for the satisfaction of integrity constraints and checks the authority of users to access data.
- It uses all the integrity constraints and authorization rules specified by the DBA.
- Integrity manager must assure data integrity during normal database operations as well as during the database failures-

Disk Storage

- A DBMS can use several kinds of data structures as a part of physical system implementation in the form of disk storage.
- Each structure has its own importance.
- Following are some common data structures.
- Disk storage is the central repository for storing all kinds of data in the database.

(i) Data

- It stores the database itself on the disk in the Data files.

(ii) Data Dictionary

- Information relating to the structure and usage of data contained in the database, the

metadata, is maintained in a data dictionary.

- The data dictionary is a database itself, documents the data.
- Each database user can consult the data dictionary to pick up what each piece of data and the various synonyms of the data fields mean.
- In a system the data dictionary is part of the DBMS (Integrated system) the data dictionary stores information concerning the source of each data-field value, the frequency of its use, and an audit trail (verification of account) concerning updates, including the who and when of each update.
- Currently data dictionary systems are available as add-ons to the DBMS.

The data dictionary stores:

- Names of relations
- Names of the attributes of each relation
- Domains, and lengths of attributes
- Names of views defined on the database, and definitions of those views
- Names of authorized users
- Accounting information about users
- Number of tuples in each relation
- Method of storage used for each relation
- Name of the index
- Name of the relation being indexed
- Attributes on which the index is defined
- Type of index formed

(iii) Indices

- Indices, which can provide fast access to data items.
- A database index provides pointers to those data items that hold a particular value.
- Hashing is an alternative to indexing that is faster in some but not all cases.

(iv) Statistical Data

- It stores statistical information about the data stored in the database, like the number of records, blocks, etc. in a table.
- This information can be used to execute a query efficiently.